

Non-drug-induced nephrotoxicity

Justine Bacchetta · Laurence Dubourg ·
Laurent Juillard · Pierre Cochat

Received: 17 January 2009 / Revised: 16 March 2009 / Accepted: 17 March 2009 / Published online: 28 April 2009
© IPNA 2009

Abstract Several drugs and other compounds can induce acute and/or chronic nephrotoxicity. The goal of this study was to review clinical features of nephrotoxicity induced by ‘atypical’ or ‘unconventional’ agents, such as environmental agents (metals, minerals, animals), food agents (mushrooms, aristolochic acid, medicinal traditional herbals, dietary supplements, melamine), drugs, and other products (ethylene glycol). Nephrotoxicity varies according to local background, dependent on different food and cultural customs, as well as to differences in local fauna and flora. The incidence of such a phenomenon is not well known. Many different pathophysiological pathways are involved, and the spectrum of renal lesions is rather wide. ‘Epidemic nephrotoxicity’ may occur, as recently illustrated by the melamine

epidemics in Chinese infants receiving powdered milk formulas; a rapid reaction to unusual increased frequency of acute kidney injury and nephrolithiasis in young children has led to a rapid analysis from international experts, with subsequent recommendations for diagnosis and care. Nephrotoxicity should be considered when there is any unexplained renal impairment, especially in children.

Keywords Aristolochic acid · Child · Ethylene glycol · Heavy metals · Melamine · Mushrooms · Nephrotoxicity

Introduction

Many different drugs and unusual agents can induce acute and/or chronic nephrotoxicity. Such nephrotoxicity may be either a dose-dependent phenomenon, following an accidental or voluntary poisoning, or a dose-independent phenomenon resulting from an immuno-allergic or vasoactive effect [1]. Although all types of renal lesions may be observed, acute tubular necrosis is the most common picture of nephrotoxicity. There is limited available information for children, so that most data come from the literature on adults and, sometimes, from experiments on animals [1]. However, there could be major differences in toxicity in children compared with that in adults, especially in infants and fetuses [2]. Because renal tubular transport capacities vary along with maturation, the degree of nephrotoxicity may also vary with growth. Moreover, as adults, predisposing factors must be taken into account in children, e.g. possible underlying nephropathy, pharmacogenomics and concomitant therapies.

Many medications are known to induce nephrotoxicity; however, ‘natural’ agents can also induce such damage. The objective of this study was to review nephrotoxicity

J. Bacchetta · P. Cochat (✉)
Centre de Référence des Maladies Rénales Rares,
Service de Néphrologie et Rhumatologie Pédiatriques,
Hôpital Femme Mère Enfant,
Boulevard Pinel,
69677 Bron, France
e-mail: pierre.cochat@chu-lyon.fr

J. Bacchetta · L. Dubourg · L. Juillard · P. Cochat
Université de Lyon,
Lyon, France

L. Dubourg
Service d’Exploration Fonctionnelle Rénale et Métabolique,
Hôpital Edouard Herriot,
Lyon, France

L. Dubourg · P. Cochat
INSERM 820,
Lyon, France

L. Juillard
Département de Néphrologie, Hôpital Edouard Herriot,
Lyon, France

induced by ‘atypical’ or ‘non-conventional’ agents, such as ethylene glycol, environmental toxins (metals, minerals, animals), food agents (mushrooms, aristolochic acid, medicinal traditional herbals, dietary supplements, melamine) and drugs (tobacco, ecstasy, marijuana, cocaine).

Pathophysiology

As 20–25% of cardiac output is directed towards the kidneys to sustain glomerular filtration and renal metabolism, renal cells can be exposed to high volumes of toxic agents [3]. The active uptake by tubular cells, the high concentration of toxins in the medullary interstitium and the presence of many different transporters and metabolizing enzymes explain the vulnerability of the kidneys to toxic agents [4]. Owing to its high metabolic activity, the proximal tubular epithelium is especially susceptible to toxic injury.

Nephrotoxicity may be secondary to acute and/or chronic exposure to toxic agents. Acute kidney injury (AKI) is generally defined as the sudden decline of glomerular filtration rate (GFR) in association with the doubling of serum creatinine level, or with a documented reduction in urine output (oliguria < 0.5 ml/kg per hour for > 6 h). Nephrotoxic drugs account for 0–7% of all AKIs, but, in children, there are few data on this topic [5]. In 1988 Bearman et al. proposed a classification of nephrotoxicity within four stages: (1) increased serum creatinine level, up to two-times the baseline; (2) increased serum creatinine level above two-times the baseline, without the requirement of renal replacement therapy; (3) AKI requiring renal replacement therapy; (4) death [6]. Drug-induced AKI can be explained by several mechanisms: pre-renal AKI, acute tubular necrosis, acute interstitial nephritis, tubular obstruction, proximal tubulopathy, rhabdomyolysis and thrombotic microangiopathy.

In vitro and in vivo experiments show that cell damage induces increased expression of many early stress response (Esr) proteins. Indeed, nephrotoxic agents can induce changes in gene and protein expression, leading to biochemical perturbations involving three main cellular pathways: intracellular calcium homeostasis, oxidative stress, and thiol modification. Moreover, a large number of drugs are known to induce renal cell apoptosis, even at low levels of drug exposure [7]. Thus, apoptosis plays a crucial role, not only during renal development, growth and remodeling, but also in various human renal diseases and drug-induced nephrotoxicity. For example, non-conventional nephrotoxic agents (e.g. cadmium, mercuric chloride and ochratoxin A) induce apoptosis in cultured tubular cells in vitro [8]. Several therapeutic strategies to limit apoptosis of tubular cells are currently being evaluated in experimental models. One strategy consists of decreasing drug accumula-

tion in the kidney via the down-regulation or inhibition of specific renal drug transporters. Another strategy aims at modulating the major pathways of apoptosis via inhibition of the pro-apoptotic Bax or up-regulation of the anti-apoptotic Bcl2, or via inhibition of caspase-dependent processes. A third strategy could be to activate growth factors to stimulate tubular regeneration [e.g. exogenous epidermal growth factor (EGF), insulin growth factor 1 (IGF1), hepatocyte growth factor (HGF), vascular endothelial growth factor (VEGF)]. These approaches are promising, especially for drug-induced nephrotoxicity (cisplatin, cidofovir, aminoglycosides, cyclosporine, amphotericin B), but they need to be evaluated for adults and for children with drug-induced nephrotoxicity and also for nephrotoxicity induced by non-conventional agents [8].

Ethylene glycol

Accidental poisoning with ethylene glycol (EG), ethane-1,2-diol, an antifreeze agent, is not unusual in children, since this colorless and odorless agent has a sweet taste and a syrupy consistency. In the context of multisystemic injury (e.g. encephalopathy, cardiovascular instability, pneumonitis), the nephrotoxicity consists of metabolic acidosis with tubular precipitation of oxalate crystals (acute oxalosis), leading to AKI 2–72 h after ingestion. In a retrospective study, metabolic acidosis was observed in 41% of patients after unintentional ingestion of EG, vomiting in 36%, nephrotoxicity in 10% and neurotoxicity in 9% [9]. EG poisoning is confirmed by the detection of EG in the patient’s blood and by quantitative analysis of glycol in the serum, but only a few laboratories are able to perform this measurement in emergency cases [10].

The addition of EG to freshly isolated proximal tubular cells fails to induce cytotoxicity, even with high doses (five- to ten-times that observed in clinical settings). Thus, EG must undergo complex metabolic conversion before it induces cytotoxic effects (mainly via alcohol dehydrogenase-mediated EG metabolism, inducing the generation of glyco-aldehyde and then glycolic, glyoxylic and oxalic acids). In animal models glyco-aldehyde and glyoxylate are the main metabolites responsible for EG nephrotoxicity [11]. The treatment of EG intoxication is symptomatic, using citrate alkalization and, sometimes, renal replacement therapy. Specific antidotes, aiming at inhibiting the production of toxic metabolites via inhibition of the alcohol dehydrogenase, can be used (ethanol or, more recently, fomepizole), without major side effects [9]. Although pyridoxine and thiamine are involved in the metabolism of glyoxylate to its non-toxic metabolites, there are no clinical data to support the clinical utility of these co-factors in the treatment of acute EG poisoning [12]. According to the American Academy of Clinical Toxicology, the criteria for antidotal therapy in EG poisoning are the

following: (1) EG concentration in serum above 200 mg/l or (2) patient's history of ingestion of a toxic dose of EG and an osmotic gap of more than 10 mosmol/kg or (3) suspicion of EG poisoning by history or clinical symptoms and two of the following criteria: arterial blood pH below 7.3, serum bicarbonate concentration below 20 mmol/l, osmotic gap above 10 mosmol/kg and crystals of calcium oxalate in urine sediment [12]. Several pediatric cases of EG poisoning with fomepizole therapy have been reported [13, 14], with similar efficacy and side effects to those in adults. Therefore, fomepizole seems to be the first choice for the treatment of EG intoxication, but it is not available in all centers [10]. Guidelines recommend hemodialysis if the EG serum level is above 500 mg/l, but the rationale for this recommendation is unclear [10].

Chronic nephrotoxicity may also be observed after an individual's exposure to glycol ether solvents (e.g. *N*-butoxy-ethanol), since they are widely used in industrial processes (e.g. for surface coating). For example, Daniel et al. reported severe glomerular mesangiolysis and peripheral neurotoxicity in a patient who, at work, had suffered long-term exposure to glycol ether solvents without protective equipment [15].

Environmental toxins: metals and minerals

Metals causing nephrotoxicity can be categorized into two different types based on their role in human physiology: those with no known function in the body (e.g. cadmium, mercury, lead, uranium) and those of which trace elements are necessary for the function of enzymes and cell proteins (e.g. copper, cobalt, iron). While these latter metals are necessary for human physiology, they can also cause nephrotoxicity when in excess [16]. Proximal tubules are the major site of metal-induced nephrotoxicity, with at least three common intracellular consequences: apoptosis, necrosis and oxidative stress.

Lead The toxic effects of lead represent the most common environmental illness among children in the United States of America. Current sources of lead are mainly paint and dust from old homes/apartments and contamination of soil by emissions from leaded fuels [17]. Infants and young children have an increased risk of lead toxicity, due to higher exposure and greater intestinal absorption relative to their body size [2]. Acute poisoning induces abdominal pain, cognitive deficits, peripheral neuropathy, arthralgia, a lead line at the junction between tooth and gum, anemia, with basophilic stippling, and renal Fanconi syndrome, due to lead accumulation in the proximal tubule; such induced Fanconi syndrome is largely reversible. By contrast, chronic poisoning will induce progressive tubular and

interstitial atrophy, with the onset of chronic kidney disease (CKD), hyperuricemia and arterial hypertension [17]. In children it can be also responsible for impaired neuro-cognitive development. It is noteworthy that the lead content of red cells only reflects recent exposure; in the case of long-term poisoning, the diagnosis must be based on an increased body burden of lead [17, 18]. As 90–95% of lead is stored in the slow turnover, calcium-dependent, skeletal pool, diagnosis and treatment depend on the extraction of lead from the tissues with ethylene diamine tetra-acetate (EDTA). Animal models of chronic lead intoxication have revealed tubular and interstitial atrophy (e.g. loss of cell brush border, cell loss, intranuclear inclusion bodies in the proximal tubule), in association with vascular lesions in small and medium-sized renal arteries (e.g. vacuolization and muscular hypertrophy of arterioles) [19]. For adults, exposure to high levels of lead (i.e. corresponding to a blood concentration above 70–80 µg/dl) is an established risk factor for CKD, and strategies aimed at decreasing the lead burden could be effective to slow down the progression of CKD [20, 21].

Cadmium Cadmium salts are increasingly used as pigments and stabilizers in plastics in industrialized countries. Children are exposed mainly from contaminated food, tobacco smoke and house dust [22]. Long-term exposure to cadmium can induce chronic nephrotoxicity and osteoporosis. Moreover, long-term exposure through inhalation during adulthood can be associated with lung cancer and other solid tumors [23].

Bismuth The use of colloidal bismuth citrate has been forbidden in France since 1974; however, in several countries, it remains a therapeutic agent for the treatment of gastroesophageal reflux. Short-term exposure to bismuth may induce AKI, whereas long-term exposure will induce toxicity to the central nervous system [24, 25]. In cases of overdose, a chelating agent (D-penicillamine) has been proposed.

Mercury Mercury is still used in gold mining and in the manufacture of batteries. Short-term exposure to this silvery white liquid metal may cause tubular necrosis, interstitial pneumonitis and gastrointestinal ulcerations. By contrast, long-term exposure can cause membranous nephropathy and central neurotoxicity [26]. A study of 49 patients with dental amalgams (of which the main component is mercury) in comparison with 52 controls demonstrated that the patients with dental amalgams had significantly higher concentrations of urinary *N*-acetyl- D-glucosaminidase (NAG) and albumin [27]. In 2008 the results of a specific pediatric trial, the New England Children's Amalgam Trial, were published: 534 children 6–10 years of age with two or

more posterior teeth caries but with no prior amalgam restorations were randomly chosen to be given amalgam or resin composites. After a 5-year follow-up period, there were no significant differences between groups for renal tubular biomarkers, but there was a significantly increased prevalence of microalbuminuria among children in the amalgam group (adjusted odds ratio 1.8, 95% confidence interval 1.1–2.9) [28].

The nephrotoxicity of other metals (or derivatives) has also been described. For example, exposure to potassium bichromate, present in the red portion of matches (and thus interesting to small children), can induce AKI [29]. Exposure to copper and uranium can cause interstitial nephritis and Fanconi syndrome [30, 31]. There is dose-related injury to proximal tubules after uranium exposure, and Magdo et al. reported in 2007 the case of a family exposed to uranium-contaminated water [32]. Only the 3-year-old boy had significant β_2 -microglobulinuria after exposure. Thus, those authors highlighted the special sensitivity of young children to residential environmental exposure, because they spend a large amount of time in their homes and because of the developmental immaturity of their kidneys and the large volume of water they consume relative to their body mass [32].

Germanium Germanium is a naturally occurring trace element widely used in the electronics industry. Several dietary supplements developed in the 1980s contained germanium, and long-term exposure to this compound has induced renal impairment (tubular damage, minor glomerular involvement) in association with extrarenal symptoms (e.g. weight loss, anorexia, anemia, muscle weakness, peripheral neuropathy, cerebellar ataxia, sensorineural impairment). Renal pathology showed foamy degeneration of tubular cells and swollen mitochondria with electron-dense particles. Death occurred in 25–33% of patients with germanium poisoning [33].

Global pollution Global pollution may also be involved in nephrotoxicity. In 2003 Kaneko et al. described general tubular dysfunction in children living near the Aral Sea, an area polluted by human activities. In their study, comparing renal function in 205 children at risk with that of 187 controls, mean urinary concentrations of NAG and β_2 -microglobulin (BMG) were both significantly higher in exposed subjects; moreover, the proportion of children with abnormal values of NAG was significantly higher in those living in a polluted area (7.9 vs 2.6%, $P < 0.05$). Those children had been exposed to various toxic chemicals (e.g. polychlorinated biphenyls, pesticides, heavy metals), secondary to intensive use of insecticides, pesticides, herbicides and defoliants, since the 1960s [34].

Silica dust exposure is associated with an increased risk of pulmonary silicosis and auto-immune disease (e.g.

progressive systemic sclerosis, rheumatoid arthritis, dermatomyositis). In terms of renal damage, it increases the risk of anti-neutrophilic cytoplasmic antibodies-associated small-vessel vasculitis, but not the risk of systemic lupus erythematosus [35, 36].

Environmental toxins: drugs

Active and passive tobacco exposure can induce acute or chronic nephrotoxicity. Short-term exposure to nicotine induces arterial hypertension and a 15% decrease in GFR, with increased synthesis of vasopressin and catecholamines [37], whereas long-term exposure induces endothelial damage and a 15% decrease in renal blood flow, without significant change in GFR [38]. Underlying nephropathies are accelerated by nicotine [39, 40]. Moreover, the increased serum levels of cadmium and lead in smokers may cause glomerular dysfunction [41]. Nicotine can also activate the sympathetic nervous system, resulting in an elevation of blood pressure, and can thus indirectly affect the kidney [42].

Ecstasy (3-4-methylene dioxy-methamphetamine) can induce rhabdomyolysis, AKI, necrotizing vasculitis, hyponatremia, acute interstitial nephritis, arterial hypertension, transient acute proximal tubular injury, and urine retention (due to bladder neck closure) [40]. Its use in the European college population ranges from 5–24% [43, 44]. Its toxicity may be secondary to increased local oxidative stress, with a dose-dependent disruption of tissue structure [45]. Early graft loss (e.g. necrotizing graft vasculopathy) from a donor with a regular consumption of ecstasy has been also reported [46].

Exposure to marijuana can induce rhabdomyolysis, renal infarction and membranous glomerulonephropathy [47, 48], whereas exposure to cocaine can induce hemodynamic and vascular changes (renal infarction, renal artery dissection), AKI and glomerulonephritis [49–52]. A case of Henoch–Schönlein purpura was described in association with cocaine consumption [40]. Intravenous injection of heroin can induce acute hypotension, acidosis, AKI, and rhabdomyolysis. Glomerulonephritis associated with bacterial, viral or fungal infections secondary to intravenous drug use has also been observed, as well as vasculitis syndromes secondary to the adulterants used to ‘cut’ intravenous drugs [40].

Environmental toxins: nephrotoxins produced by animals

Snake bite nephropathy accounts worldwide for 125,000 deaths (out of 5,000,000 poisonous snake bites) yearly, according to the World Health Organization [53]. Enveno-

mation is, therefore, a frequent cause of AKI in children in developing countries [54]. It can induce moderate renal lesions (mild proteinuria, hematuria) but also more severe damage: acute tubular necrosis, patchy or diffuse cortical necrosis, thrombotic microangiopathy and mesangiolysis, intravascular hemolysis, rhabdomyolysis, vasculitis, thus leading to oliguric or non-oliguric AKI [54]. The incidence of renal involvement after snake bite varies from 1.4–28% [55]. The commonest species responsible for renal involvement are Russell's viper (*Echis carinatus*), the puffadder and the sea snake. In a prospective study of 100 patients in Brazil in 2005 after envenomation by *Crotalus* spp., the incidence of AKI, defined by a decrease in GFR below 60 ml/min per 1.73 m² during the first 72 h after envenomation, was 29%; 24% of patients needed renal replacement therapy and 10% died [55]. The risk factors for the onset of AKI were a smaller body surface area, an age below 12 years, a marked increase in creatine kinase concentration (> 2,000 IU/l), and a longer delay in anti-venom treatment. Diuresis at admission of above 90 ml/h was an independent protection factor for AKI [55]. Treatment of envenomation is symptomatic (renal replacement therapy or plasmapheresis when required); in several cases, specific anti-venom treatments are available. Monovalent anti-venom agents are preferred and should be given before the onset of AKI [54].

Nephrotoxicity following a bite or sting by small animals (e.g. spiders, arthropods, scorpions, hymenoptera, caterpillars) can induce allergic and systemic symptoms, but also various types of nephrotoxicity such as AKI, acute tubular necrosis, interstitial nephritis, hemolytic uremic syndrome, nephrotic syndrome and, more rarely, pigment tubulopathy [56–60]. Marine animals (e.g. carp, jellyfish, sea anemones) can also induce acute tubular necrosis, AKI, proteinuria and nephrotic syndrome [61–64].

Deforestation Deforestation and elimination of natural predators in several countries have modified the global ecosystem, inducing major ecological consequences but also an increased frequency of accidental poisoning by animal toxins. This was recently described by Riella et al. [65]. For example, poisoning by the caterpillar *Lonomia obliqua* can induce AKI and severe coagulopathy, and the mortality rate in Brazil peaked at 20% in the 1990s before falling to <2% following the introduction of a specific antiserum in 1995 and educational campaigns. The recent death of a young Canadian patient highlights the importance of considering this diagnosis in patients traveling back from South America (mainly Brazil and Venezuela), especially when symptoms occur after a symptom-free interval period [66]. Renal lesions appear to be secondary to massive deposition of fibrin in glomeruli, together with direct endothelial and tubular injury [65].

Food nephrotoxicity: mushrooms and fish

Orellanine, a tetrahydroxy-bipyridine-disoxide toxin, is derived from *Cortinarius* mushroom spp., mainly *Cortinarius orellanus* and *Cortinarius speciosissimus*. Such mushrooms usually grow in semi-mountainous forests at the end of summertime and the beginning of fall. Orellanine is resistant to heat, freezing, drying and cooking. Intoxication can induce gastrointestinal symptoms (median onset of symptoms 3 days after ingestion), loin pain and delayed acute tubulointerstitial nephritis, with a symptom-free interval (median onset 9 days after ingestion; but renal symptoms can occur 2–21 days after ingestion) [67]. Renal histopathological examination can show interstitial edema and lymphoplasmacytic infiltration. Hepatotoxicity and muscle lesions may also be present. The diagnosis can be established by mycologic analysis or by identification of the toxin in renal tissue. Orellanine in the serum can also be detected by chromatography. It is noteworthy that urinary excretion of the toxin lasts only for approximately 24 h, so that negative results of a urine test cannot rule out such a diagnosis. The treatment is supportive, and therapeutic options include renal replacement therapy, plasmapheresis, diltiazem administration, and corticosteroid and antioxidant therapy (*N*-acetylcysteine), with varying results. However, hemodialysis and early plasma exchange do not seem to be very effective, probably because of the symptom-free interval between ingestion and the occurrence of renal symptoms [68]. A systematic review (*n*=245 patients) showed that, despite such therapies, AKI progressed towards CKD in 50% of cases; among them, CKD led to end-stage renal disease in 70% of cases [67].

Ochratoxin A Ochratoxin A (OTA) is a mycotoxin produced mainly by *Aspergillus ochraceus* and *Penicillium verrucosum*, which can be present in human milk. In an Egyptian study of 50 healthy breast-fed infants in comparison with 36 breast-fed children exposed to OTA, univariate analysis showed that the presence of OTA was associated with significantly higher urinary levels of albumin and BMG. In a multivariate logistic regression analysis, there was a significant correlation between a higher serum OTA concentration and the degree of urinary microalbumin [69].

Other types of mushrooms can induce nephrotoxicity: acute tubulopathy from *Amanita* spp., and rhabdomyolysis from the *Tricholoma equestre* mushroom [70, 71].

Another food that can cause nephrotoxicity is fish gallbladders, which are used in Asian traditional medicine. They can induce acute tubular necrosis and AKI. The toxin involved is thought to be cyprinol sulfate, which can induce gastrointestinal, renal, hepatic, cardiac and neurological injuries [72].

Food nephrotoxicity: Chinese herbs and aristolochic acids

Dietary use of aristolochic acid ($C_{17}H_{11}NO_7$), a toxin derived from *Aristolochia* plant spp. that have carcinogenic and mutagenic properties, can cause several different clinical syndromes. The clinical picture can vary from aristolochic acid nephropathy (AAN) in Western Europe, to endemic Balkan nephropathy (EBN) in Eastern Europe, and to reversible renal Fanconi syndrome in Asian countries [73–75].

AAN was first described in Belgium in the early 1990s when it affected women ingesting aristolochic acid in weight-reducing pills [76]. In such cases, the aristolochic acid came from the *Aristolochia fangchi* plant. AAN has rarely been reported in children [77].

EBN affects subjects living in rural areas of Eastern Europe. Its onset is insidious and leads to progressive CKD. EBN has recently been shown to be caused by long-term dietary poisoning by aristolochic acid from *Aristolochia clematitis* [75].

AAN and EBN present common features, e.g. tubular proteinuria, proximal tubulopathy, aseptic leukocyturia, early onset of severe anemia and decreased kidney size. In both conditions renal biopsy shows hypocellular interstitial fibrosis, tubular atrophy, limited interstitial infiltrates and mild glomerular damage. Aristolochic acid can also lead to urothelial malignancies. However, the progression to end-stage renal disease is rather slow in EBN (within approximately 20 years), whereas it is rapid in AAN (between 6 months and 2 years) [78].

Such regional discrepancies could be explained by differences in the amount of aristolochic acid ingested (and the amount ingested within a given time period), by different genetic backgrounds, and by the differences among various so-called aristolochic acids. Traditional Chinese medicines are often a mixture of several herbs, and aristolochic acid could come from at least six different herbs, leading to the concept of ‘mixed nephropathies’ rather than AAN [79].

The pathophysiology of aristolochic acid nephrotoxicity has been studied in rabbits, which have renal lesions similar to those observed in humans [80]. Aristolochic acid seems to have a direct nephrotoxic effect on tubular cells, but it also induces apoptosis and fibrosis (via increased production of extracellular matrix components and profibrotic growth factors) [81].

Other medicinal plants and dietary supplements

Complementary and alternative medicines The prescription of complementary and alternative medicines has gradually increased worldwide, whereas systematic reviews available

on the efficacy of herbal preparations have shown very little evidence of effectiveness of ‘herbalism’ and do not provide any data on side effects associated with such complementary therapies. According to a study performed in 1999 by the Center for Disease Control, 10% of American adults regularly ingest herbal medicines [82]. Other studies report that almost half of the American population uses some form of complementary and alternative medicine [83]. In developing countries the prevalence of nephropathies caused by traditional medicines results from a combination of poverty, lack of medical facilities, absence of specific legislation and belief in ancestral remedies [84]. In Africa, the use of traditional herbal remedies has been reported to account for 35% of all cases of AKI [85].

At least four categories of nephrotoxicity secondary to ingestion of medicinal herbs can be distinguished: (1) the plant is correctly identified, but its renal toxicity is unknown or underestimated; (2) there is an accidental or a deliberate contamination in the preparation of the herbal mixture with drugs, hormones or heavy metals; (3) the herb has been incorrectly identified; and (4) there is an interaction between the product and conventional therapies [79, 86].

Traditional medicinal products can induce various renal problems: tubular necrosis, acute or chronic interstitial nephritis, Fanconi syndrome, dyskalemia, arterial hypertension, papillary necrosis, urolithiasis and urothelial cancer. For example, products containing *Glycyrrhiza* spp. may induce arterial hypertension. Cranberry juice, due to its high oxalate content, may lead to nephrolithiasis [86].

In Morocco ‘takaout roumia’, or para-phenylene diamine (PPD), is a black pigment used as a cosmetic product. In acute self-poisoning, the urine becomes dark, and rhabdomyolysis/AKI can occur. Temporary intubation may also be necessary, as there is often upper-airway angioneurotic edema. In Morocco PPD-induced AKI accounts for 10% of patients with acute tubular necrosis, 50% of all cases of rhabdomyolysis and 66% of deaths from toxins [87].

In patients with underlying nephropathy and CKD, the use of herbal preparations can be problematic, due to their high potassium concentration, their ability to interact with conventional drugs and/or their vasoconstrictive properties. For example, the risk of decreased serum levels of calcineurin inhibitors is increased by *Hypericum perforatum*, which modifies cytochrome P-450 CYP3A4 activity [86, 88].

Dietary supplements and high-protein diets Dietary supplements can also cause nephrotoxicity. To date, at least 17 dietary supplements have been associated with direct renal injury [83]. Moreover, high-protein diets are very popular, but they may have renal consequences: large epidemiological studies have found an association between high dietary protein intake and both the presence of microalbuminuria and the worsening of renal function [89, 90]. Such

nephrotoxicity may likely occur via enhanced production of advanced glycation end-products (AGEs), secondary to increased amino acid load. These dietary protein-derived AGEs may contribute to pro-inflammatory and pro-oxidative processes in diabetes and CKD, but further studies are needed to explore this hypothesis completely [91].

New food-related toxicity: the case of melamine intoxication in powdered milk formulas

Melamine (melamine formaldehyde, $C_3H_6N_6$, or cyanuramide) is widely used industrially as a plastic resin (products with resin-based coatings). The use of melamine as a food adjuvant is strictly forbidden in many countries. However, because of its high nitrogen content, manufacturers have illegally added it to food to trick food testers into thinking there is a higher protein content. In September 2008, the United States Food and Drug Administration learned that a large outbreak of kidney stones (over 54 000 cases at the time of this report) and sometimes AKI and death (at least three confirmed cases) had been reported in Chinese children, due to the abnormal presence of melamine in powdered milk formulas. There were over 14,000 hospitalizations. These children (82% of them were younger than 2 years) presented with nephrolithiasis and AKI. Kidney stones were analyzed and revealed dehydrated uric acid and triple ammonium phosphates. The stones were not uniformly radio-opaque but they were visible by ultrasound or computed tomography. The following symptoms were reported: unexplained crying (especially when voiding), vomiting, hematuria, acute obstructive renal failure, stones, fever due to urinary tract infection, arterial hypertension, tenderness and flank pain. A few patients developed

ascites [92]. In experiments on mice and rats, long-term exposure to melamine has been shown to lead to kidney and bladder stones and, possibly, transitional cell carcinoma of the bladder [93]. Thompson et al. studied kidneys from three dogs that developed AKI after eating canned pet food with the addition of melamine [94]. There was acute and marked tubular degeneration and necrosis, with lymphoplasmacytic interstitial infiltrate and many intratubular birefringent crystals [95]. Two types of crystals were identified: melamine-containing crystals and calcium oxalate crystals. Overall stone formation seems to be dose-, threshold- and concentration-dependent.

The pathophysiology of such nephrotoxicity in humans is not well known, but cyanuric acid, a metabolite of melamine, appears to play a key role in the genesis of stones and AKI. As melamine becomes undetectable in rat tissues 24 h after exposure, nephrotoxicity may be related to acute overload rather than to progressive cumulative poisoning.

Conclusion

The nephrotoxicity of non-conventional agents, the incidence of which is not well known, may be either a short or long-term phenomenon and may involve different pathophysiological pathways, as summarized in Table 1. Nephrotoxicity can vary worldwide, according to dietary and cultural background and to differences in the local fauna and flora; however, local or larger epidemics have occurred, as illustrated by the melamine epidemic in China in 2008 and by the epidemics of AKI and death in Moroccan infants due to harmel exposure in 2000 (Cochat P, unpublished data). During these two episodes, prompt reaction to unusually increased

Table 1 Nephrotoxicity from non-conventional nephrotoxic agents

Site affected/mechanism	Non-conventional nephrotoxic agent (short-term toxicity)	Non-conventional nephrotoxic agent (long-term toxicity)
Prerenal failure/hemodynamic alteration	Cocaine; animals	
Kidney failure	Ethylene glycol; potassium bichromate; animals; mushrooms; melamine; creatine; licorice	<i>Cortinarius</i> intoxication
Glomerular damage/proteinuria	Mercury; animals	Mercury; gold; bismuth; glycol ether solvents
Hemolytic and uremic syndrome/thrombotic microangiopathy	Animals	
Proximal tubular injury (Fanconi syndrome, complete or not)	Lead; cadmium; mercuric chloride; aristolochic acid; Ecstasy; L-lysine	Heavy metals; L-lysine
Interstitial nephritis	Traditional herbal medicines; Cat's claw; creatine	Aristolochic acid; lead; L-lysine
Tubular necrosis (acute) or degeneration (chronic)	Heavy metals (mercury); potassium bichromate; animals; fish gallbladders; traditional herbal medicines; chromium; Yellow oleander	Germanium, Pennyroyal
Crystallization/nephrolithiasis	Ethylene glycol; melamine; silicate-containing milk thickener; star fruit	Cranberry; ephedra; vitamin C
Immune vasculitis/lupus		Silica; yohimbe

frequency of AKI in children led to rapid investigation by international experts, leading to improvements in diagnosis, care and prevention. Nephrotoxicity must, therefore, be considered in cases of unexplained renal dysfunction, especially in children.

Treatment of non-conventional agent-induced nephrotoxicity is mainly symptomatic, but it may require renal replacement therapy. The mortality rate of children with AKI ranges from 8–89% in retrospective series and from 30–36% in prospective studies. Askenazi et al. showed a 5-year survival rate of 80% in children with AKI from varying causes (among them 16% by nephrotoxins) surviving the initial period, but an overall survival rate of 57% [96]. It is interesting to note that, among the 29 children examined 3–5 years after the acute episode, 59% of them had clinical or biological symptoms of renal sequelae (arterial hypertension, decreased GFR, microalbuminuria) [97]. Thus, children who are recovering and have ‘normal’ renal function after toxin-induced AKI should be followed up regularly.

In the future systematic reports of previously undescribed non-conventional nephrotoxicity episodes and pediatric studies evaluating specific effects of nephrotoxins on the developing kidney may help to improve the understanding of the underlying pathophysiology and the care of such children.

References

1. Cochat P, Bacchetta J (2008) Néphrotoxicité. *EMC Pédiatrie*. 4-084-B-10
2. Solhaug MJ, Bolger PM, Jose PA (2004) The developing kidney and environmental toxins. *Pediatrics* 113:1084–1091
3. Ferguson MA, Vaidya VS, Bonventre JV (2008) Biomarkers of nephrotoxic acute kidney injury. *Toxicology* 245:182–193
4. Van Vleet TR, Schnellmann RG (2003) Toxic nephropathy: environmental chemicals. *Semin Nephrol* 23:500–508
5. Patzer L (2008) Nephrotoxicity as a cause of acute kidney injury in children. *Pediatr Nephrol* 23:2159–2173
6. Bearman SI, Appelbaum FR, Buckner CD, Petersen FB, Fisher LD, Clift RA, Thomas ED (1988) Regimen-related toxicity in patients undergoing bone marrow transplantation. *J Clin Oncol* 6:1562–1568
7. van de Water B, de Graauw M, Le Devedec S, Alderliesten M (2006) Cellular stress responses and molecular mechanisms of nephrotoxicity. *Toxicol Lett* 162:83–93
8. Servais H, Ortiz A, Devuyst O, Denamur S, Tulkens PM, Mingot-Leclercq MP (2008) Renal cell apoptosis induced by nephrotoxic drugs: cellular and molecular mechanisms and potential approaches to modulation. *Apoptosis* 13:11–32
9. Krenova M, Pelclova D (2007) Does unintentional ingestion of ethylene glycol represent a serious risk? *Hum Exp Toxicol* 26:59–67
10. Schwerk N, Desel H, Schulz M, Schwerk C, Kiess W, Siekmeyer W (2007) Successful therapy of paediatric ethylene glycol poisoning: a case report and annual survey by a regional poison centre. *Acta Paediatr* 96:461–463
11. Poldelski V, Johnson A, Wright S, Rosa VD, Zager RA (2001) Ethylene glycol-mediated tubular injury: identification of critical metabolites and injury pathways. *Am J Kidney Dis* 38:339–348
12. Barceloux DG, Krenzelok EP, Olson K, Watson W (1999) American Academy of Clinical Toxicology Practice Guidelines on the Treatment of Ethylene Glycol Poisoning. Ad Hoc Committee. *J Toxicol Clin Toxicol* 37:537–560
13. Caravati EM, Heilesen HL, Jones M (2004) Treatment of severe pediatric ethylene glycol intoxication without hemodialysis. *J Toxicol Clin Toxicol* 42:255–259
14. Harry P, Jobard E, Briand M, Caubet A, Turcant A (1998) Ethylene glycol poisoning in a child treated with 4-methylpyrazole. *Pediatrics* 102:E31
15. Daniel L, Robert A, Lesavre P, Figarella-Branger D (2004) Severe mesangiolysis in a patient exposed to glycol ether solvents. *Nephrol Dial Transplant* 19:2679
16. Sabolic I (2006) Common mechanisms in nephropathy induced by toxic metals. *Nephron Physiol* 104:p107–114
17. Marsden PA (2003) Increased body lead burden—cause or consequence of chronic renal insufficiency? *N Engl J Med* 348:345–347
18. Fontanellas A, Navarro S, Moran-Jimenez MJ, Sanchez-Fructuoso AI, Vegh I, Barrientos A, de Salamanca RE (2002) Erythrocyte aminolevulinatase activity as a lead marker in patients with chronic renal failure. *Am J Kidney Dis* 40:43–50
19. Sanchez-Fructuoso AI, Blanco J, Cano M, Ortega L, Arroyo M, Fernandez C, Prats D, Barrientos A (2002) Experimental lead nephropathy: treatment with calcium disodium ethylenediaminetetraacetate. *Am J Kidney Dis* 40:59–67
20. Ekong EB, Jaar BG, Weaver VM (2006) Lead-related nephrotoxicity: a review of the epidemiologic evidence. *Kidney Int* 70:2074–2084
21. Lin JL, Lin-Tan DT, Hsu KH, Yu CC (2003) Environmental lead exposure and progression of chronic renal diseases in patients without diabetes. *N Engl J Med* 348:277–286
22. Schoeters G, Den Hond E, Zuurbier M, Naginiene R, van den Hazel P, Stilianakis N, Ronchetti R, Koppe JG (2006) Cadmium and children: exposure and health effects. *Acta Paediatr Suppl* 95:50–54
23. Huff J, Lunn RM, Waalkes MP, Tomatis L, Infante PF (2007) Cadmium-induced cancers in animals and in humans. *Int J Occup Environ Health* 13:202–212
24. Cengiz N, Uslu Y, Gok F, Anarat A (2005) Acute renal failure after overdose of colloidal bismuth subcitrate. *Pediatr Nephrol* 20:1355–1358
25. Islek I, Uysal S, Gok F, Dundaroz R, Kucukoduk S (2001) Reversible nephrotoxicity after overdose of colloidal bismuth subcitrate. *Pediatr Nephrol* 16:510–514
26. Aymaz S, Gross O, Krakamp B, Ortmann M, Dienes HP, Weber M (2001) Membranous nephropathy from exposure to mercury in the fluorescent-tube-recycling industry. *Nephrol Dial Transplant* 16:2253–2255
27. Mortada WL, Sobh MA, El-Defrawy MM, Farahat SE (2002) Mercury in dental restoration: is there a risk of nephrotoxicity? *J Nephrol* 15:171–176
28. Barregard L, Trachtenberg F, McKinlay S (2008) Renal effects of dental amalgam in children: the New England children’s amalgam trial. *Environ Health Perspect* 116:394–399
29. Picaud JC, Cochat P, Parchoux B, Berthier JC, Gilly J, Chareyre S, Larbre F (1991) Acute renal failure in a child after chewing of match heads. *Nephron* 57:225–226
30. Bhowmik D, Mathur R, Bhargava Y, Dinda AK, Agarwal SK, Tiwari SC, Dash SC (2001) Chronic interstitial nephritis following parenteral copper sulfate poisoning. *Ren Fail* 23:731–735
31. Zimmerman KL, Barber DS, Ehrich MF, Tobias L, Hancock S, Hinckley J, Binder EM, Jortner BS (2007) Temporal clinical chemistry and microscopic renal effects following acute uranyl acetate exposure. *Toxicol Pathol* 35:1000–1009
32. Magdo HS, Forman J, Graber N, Newman B, Klein K, Satlin L, Amler RW, Winston JA, Landrigan PJ (2007) Grand rounds:

- nephrotoxicity in a young child exposed to uranium from contaminated well water. *Environ Health Perspect* 115:1237–1241
33. Tao SH, Bolger PM (1997) Hazard assessment of germanium supplements. *Regul Toxicol Pharmacol* 25:211–219
34. Kaneko K, Chiba M, Hashizume M, Kunii O, Sasaki S, Shimoda T, Yamashiro Y, Caypil W, Dauletbaev D (2003) Renal tubular dysfunction in children living in the Aral Sea Region. *Arch Dis Child* 88:966–968
35. Hogan SL, Satterly KK, Dooley MA, Nachman PH, Jennette JC, Falk RJ (2001) Silica exposure in anti-neutrophil cytoplasmic autoantibody-associated glomerulonephritis and lupus nephritis. *J Am Soc Nephrol* 12:134–142
36. Stratta P, Canavese C, Messuerotti A, Fenoglio I, Fubini B (2001) Silica and renal diseases: no longer a problem in the 21st century? *J Nephrol* 14:228–247
37. Simon P, Halimi JM, Cordonnier D (2001) Tobacco and kidney. *Nephrologie* 22:137–140
38. Gambaro G, Verlato F, Budakovic A, Casara D, Saladini G, Del Prete D, Bertaglia G, Masiero M, Checchetto S, Baggio B (1998) Renal impairment in chronic cigarette smokers. *J Am Soc Nephrol* 9:562–567
39. Cooper RG (2006) Effect of tobacco smoking on renal function. *Indian J Med Res* 124:261–268
40. Crowe AV, Howse M, Bell GM, Henry JA (2000) Substance abuse and the kidney. *QJM* 93:147–152
41. Satarug S, Ujjin P, Vanavanitkun Y, Nishijo M, Baker JR, Moore MR (2004) Effects of cigarette smoking and exposure to cadmium and lead on phenotypic variability of hepatic CYP2A6 and renal function biomarkers in men. *Toxicology* 204:161–173
42. Halimi JM, Philippon C, Mimran A (1998) Contrasting renal effects of nicotine in smokers and non-smokers. *Nephrol Dial Transplant* 13:940–944
43. Kwon C, Zaritsky A, Dharmidharka VR (2003) Transient proximal tubular renal injury following Ecstasy ingestion. *Pediatr Nephrol* 18:820–822
44. Woodrow G, Turney JH (1999) Ecstasy-induced renal vasculitis. *Nephrol Dial Transplant* 14:798
45. Ninkovic M, Selakovic V, Dukic M, Milosavljevic P, Vasiljevic I, Jovanovic M, Malicevic Z (2008) Oxidative stress in rat kidneys due to 3, 4-methylenedioxymetamphetamine (ecstasy) toxicity. *Nephrology (Carlton)* 13:33–37
46. Hurault de Ligny B, El Haggan W, Comoz F, Lobbedez T, Pujo M, Griveau AM, Bottet P, Bensadoun H, Ryckelynck JP (2005) Early loss of two renal grafts obtained from the same donor: role of ecstasy? *Transplantation* 80:153–156
47. Farber SJ, Huertas VE (1976) Intravenously injected marihuana syndrome. *Arch Intern Med* 136:337–339
48. Bohatyrewicz M, Urasinska E, Rozanski J, Ciechanowski K (2007) Membranous glomerulonephritis may be associated with heavy marijuana abuse. *Transplant Proc* 39:3054–3056
49. Glauser J, Queen JR (2007) An overview of non-cardiac cocaine toxicity. *J Emerg Med* 32:181–186
50. Bermanian S, Motallebi M, Nosrati SM (2005) Cocaine-induced renal infarction: report of a case and review of the literature. *BMC Nephrol* 6:10
51. Edmondson DA, Towne JB, Foley DW, Abu-Hajir M, Kochar MS (2004) Cocaine-induced renal artery dissection and thrombosis leading to renal infarction. *WMJ* 103:66–69
52. Nzerue CM, Hewan-Lowe K, Riley LJ Jr (2000) Cocaine and the kidney: a synthesis of pathophysiologic and clinical perspectives. *Am J Kidney Dis* 35:783–795
53. Chippaux JP (1998) Snake-bites: appraisal of the global situation. *Bull World Health Organ* 76:515–524
54. Sitprija V (2006) Snakebite nephropathy. *Nephrology (Carlton)* 11:442–448
55. Pinho FM, Zanetta DM, Burdmann EA (2005) Acute renal failure after *Crotalus durissus* snakebite: a prospective survey on 100 patients. *Kidney Int* 67:659–667
56. Reisman RE (2005) Unusual reactions to insect stings. *Curr Opin Allergy Clin Immunol* 5:355–358
57. Hasan S, Hassan K (2005) Proteinuria associated with centipede bite. *Pediatr Nephrol* 20:550–551
58. Chao YW, Yang AH, Ng YY, Yang WC (2004) Acute interstitial nephritis and pigmented tubulopathy in a patient after wasp stings. *Am J Kidney Dis* 43:e15–19
59. Bhatta N, Singh R, Sharma S, Sinnha A, Raja S (2005) Acute renal failure following multiple wasp stings. *Pediatr Nephrol* 20:1809–1810
60. Bahloul M, Ben Hmida M, Belhouli W, Ksibi H, Kallel H, Ben Hamida C, Chaari A, Chelly H, Rekik N, Bouaziz M (2004) Le syndrome hémolytique et urémique secondaire à une envenimation scorpionique: à propos de 2 cas. *Néphrologie* 25:49–51
61. Deekajorndech T, Kingwatanakul P, Wananukul S (2004) Acute renal failure in a child with jellyfish contact dermatitis. *J Med Assoc Thai* 87(Suppl 2):S292–294
62. Mizuno M, Nozaki M, Morine N, Suzuki N, Nishikawa K, Morgan BP, Matsuo S (2007) A protein toxin from the sea anemone *Phyllodiscus semoni* targets the kidney and causes a severe renal injury with predominant glomerular endothelial damage. *Am J Pathol* 171:402–414
63. Mizuno M, Nishikawa K, Yuzawa Y, Kanie T, Mori H, Araki Y, Hotta N, Matsuo S (2000) Acute renal failure after a sea anemone sting. *Am J Kidney Dis* 36:E10
64. Cochat P, Larbre F (1988) Nephrotic syndrome after a medusa sting. *Pediatrue* 43:419–420
65. Riella MC, Chula D, de Freitas S, Mazza M, Pachaly M (2008) Acute renal failure and haemorrhagic syndrome secondary to toxin of caterpillars (*Lonomia obliqua*). *NDT Plus* 6:445–446
66. Chan K, Lee A, Onell R, Etches W, Nahiriak S, Bagshaw SM, Larratt LM (2008) Caterpillar-induced bleeding syndrome in a returning traveller. *CMAJ* 179:158–161
67. Saviuc P, Garon D, Danel V, Richard JM (2001) Cortinari poisoning. Analysis of cases in the literature. *Nephrologie* 22:167–173
68. Wornle M, Angstwurm MW, Sitter T (2004) Treatment of intoxication with Cortinari species using an antioxidant therapy. *Am J Kidney Dis* 43:e3–6
69. Hassan AM, Sheashaa HA, Fattah MF, Ibrahim AZ, Gaber OA, Sobh MA (2006) Study of ochratoxin A as an environmental risk that causes renal injury in breast-fed Egyptian infants. *Pediatr Nephrol* 21:102–105
70. Saviuc P, Danel V (2006) New syndromes in mushroom poisoning. *Toxicol Rev* 25:199–209
71. Bedry R, Baudrimont I, Deffieux G, Creppy EE, Pomies JP, Ragnaud JM, Dupon M, Neau D, Gabinski C, De Witte S, Chapalain JC, Godeau P, Beylot J (2001) Wild-mushroom intoxication as a cause of rhabdomyolysis. *N Engl J Med* 345:798–802
72. Xuan BH, Thi TX, Nguyen ST, Goldfarb DS, Stokes MB, Rabenou RA (2003) Ichthyotoxic ARF after fish gallbladder ingestion: a large case series from Vietnam. *Am J Kidney Dis* 41:220–224
73. Debelle FD, Vanherweghem JL, Nortier JL (2008) Aristolochic acid nephropathy: a worldwide problem. *Kidney Int* 74:158–169
74. Lee S, Lee T, Lee B, Choi H, Yang M, Ihm CG, Kim M (2004) Fanconi's syndrome and subsequent progressive renal failure caused by a Chinese herb containing aristolochic acid. *Nephrology (Carlton)* 9:126–129
75. Grollman AP, Shibutani S, Moriya M, Miller F, Wu L, Moll U, Suzuki N, Fernandes A, Rosenquist T, Medverec Z, Jakovina K, Brdar B, Slade N, Turesky RJ, Goodenough AK, Rieger R, Vukelic M, Jelakovic B (2007) Aristolochic acid and the etiology of endemic (Balkan) nephropathy. *Proc Natl Acad Sci U S A* 104:12129–12134

76. Debelle F, Nortier J (2006) Nephropathy caused by Chinese plants and aristolochic acids: from clinical observation to experimental model. *Bull Mem Acad R Med Belg* 161:327–333
77. Hong YT, Fu LS, Chung LH, Hung SC, Huang YT, Chi CS (2006) Fanconi's syndrome, interstitial fibrosis and renal failure by aristolochic acid in Chinese herbs. *Pediatr Nephrol* 21:577–579
78. de Jonge H, Vanrenterghem Y (2008) Aristolochic acid: the common culprit of Chinese herbs nephropathy and Balkan endemic nephropathy. *Nephrol Dial Transplant* 23:39–41
79. Nortier JL, Vanherweghem JL (2007) For patients taking herbal therapy—lessons from aristolochic acid nephropathy. *Nephrol Dial Transplant* 22:1512–1517
80. Cosyns JP, Dehoux JP, Guiot Y, Goebbels RM, Robert A, Bernard AM, van Ypersele de Strihou C (2001) Chronic aristolochic acid toxicity in rabbits: a model of Chinese herbs nephropathy? *Kidney Int* 59:2164–2173
81. Li X, Wang H (2004) Aristolochic acid nephropathy: what we know and what we have to do. *Nephrology (Carlton)* 9:109–111
82. Barrett B, Kiefer D, Rabago D (1999) Assessing the risks and benefits of herbal medicine: an overview of scientific evidence. *Altern Ther Health Med* 5:40–49
83. Gabardi S, Munz K, Ulbricht C (2007) A review of dietary supplement-induced renal dysfunction. *Clin J Am Soc Nephrol* 2:757–765
84. Jha V, Chugh KS (2003) Nephropathy associated with animal, plant, and chemical toxins in the tropics. *Semin Nephrol* 23:49–65
85. Adekun TA, Ekwere TR, Akinsola A (1999) The pattern of acute toxic nephropathy in Ife, Nigeria. *West Afr J Med* 18:60–63
86. Isnard Bagnis C, Deray G, Baumelou A, Le Quintrec M, Vanherweghem JL (2004) Herbs and the kidney. *Am J Kidney Dis* 44:1–11
87. Bourquia A, Jabrane AJ, Ramdani B, Zaid D (1988) Systemic toxicity of paraphenylenediamine. 4 cases. *Presse Med* 17:1798–1800
88. Bolley R, Zulke C, Kammerl M, Fischereder M, Kramer BK (2002) Tacrolimus-induced nephrotoxicity unmasked by induction of the CYP3A4 system with St John's wort. *Transplantation* 73:1009
89. Knight EL, Stampfer MJ, Hankinson SE, Spiegelman D, Curhan GC (2003) The impact of protein intake on renal function decline in women with normal renal function or mild renal insufficiency. *Ann Intern Med* 138:460–467
90. Hoogeveen EK, Kostense PJ, Jager A, Heine RJ, Jakobs C, Bouter LM, Donker AJ, Stehouwer CD (1998) Serum homocysteine level and protein intake are related to risk of microalbuminuria: the Hoorn Study. *Kidney Int* 54:203–209
91. Uribarri J, Tuttle KR (2006) Advanced glycation end products and nephrotoxicity of high-protein diets. *Clin J Am Soc Nephrol* 1:1293–1299
92. Langman CB, Alon U, Ingelfinger J, Englund M, Saland JM, Somers MJ, Stapleton FB, Sibbald NO, Cochat P, Wong W, Eke FU, Satlin L, Salusky I (2009) A position statement on kidney disease from powdered infant formula-based melamine exposure in Chinese infants. *Pediatr Nephrol* doi:10.1007/s00467-009-1129-6
93. Melnick RL, Boorman GA, Haseman JK, Montali RJ, Huff J (1984) Urolithiasis and bladder carcinogenicity of melamine in rodents. *Toxicol Appl Pharmacol* 72:292–303
94. Thompson ME, Lewin-Smith MR, Kalasinsky VF, Pizzolato KM, Fleetwood ML, McElhaney MR, Johnson TO (2008) Characterization of melamine-containing and calcium oxalate crystals in three dogs with suspected pet food-induced nephrotoxicosis. *Vet Pathol* 45:417–426
95. Cianciolo RE, Bischoff K, Ebel JG, Van Winkle TJ, Goldstein RE, Serfilippi LM (2008) Clinicopathologic, histologic, and toxicologic findings in 70 cats inadvertently exposed to pet food contaminated with melamine and cyanuric acid. *J Am Vet Med Assoc* 233:729–737
96. Hui-Stickle S, Brewer ED, Goldstein SL (2005) Pediatric ARF epidemiology at a tertiary care center from 1999 to 2001. *Am J Kidney Dis* 45:96–101
97. Askenazi DJ, Feig DI, Graham NM, Hui-Stickle S, Goldstein SL (2006) 3–5 year longitudinal follow-up of pediatric patients after acute renal failure. *Kidney Int* 69:184–189